PayPal-AgentFlow

Autonomous Multi-Agent Commerce & Escrow Protocol

- Submission: PayPal AI Hackathon 2026 (Devpost)
- Architects: Mushfiqur Rahim & L
- Core Tech: PayPal Orders v2 API + ReAct Multi-Agent Consensus + FastAPI
- Vision: Enabling autonomous AI agents to negotiate, invoice, and settle payments with zero friction.

The Problem: Agent-to-Agent Commerce Friction

Why Traditional Checkouts Break in the Autonomous Agent Era

- Manual Invoice Bottlenecks: Freelancers and AI services lose hours generating manual invoices.
- Dispute & Fraud Risk: Freelance milestones suffer from chargeback fraud and payment delays.
- Agentic Economy Deficit: Modern LLMs can write code, but cannot legally or safely hold and release funds.
- Need for Protocol: A secure bridge connecting autonomous agents with regulated banking rails (PayPal).

The Solution: PayPal Orders v2 Integration

Tri-Agent Autonomous Workflow with Deterministic Risk Auditing

- Negotiator Agent: Parses natural language statements into formal milestone agreements.
- RiskAudit Agent: Evaluates counterparty trust score (0-100) and checks transaction velocity.
- Settlement Agent: Interacts with PayPal Orders v2 API to create capture-intent orders in seconds.
- PayPal Smart SDK: Renders official PayPal buttons directly in the operator HUD.

Interactive HUD & Developer Experience

Zero-Friction Live Demo Deployed Globally on GitHub Pages

- Standalone WebGL/Canvas HUD: Live transaction stream, risk telemetry, and PayPal SDK buttons.
- Zero-Friction Sandbox Fallback: Judges can test full checkout flows immediately without credentials.
- Deterministic Latency: Sub-10ms pipeline execution from prompt to checkout button.
- Modular REST API: Easily integrated into any existing e-commerce or SaaS platform.

Roadmap & Production Expansion

From Hackathon Prototype to Global AI Commerce Standard

- Phase 1 (Complete): Orders v2 Sandbox client, Tri-agent consensus, interactive HUD, test suite.
- Phase 2: PayPal Payouts API integration for automated contractor micro-distributions.
- Phase 3: Webhook event verification for automated escrow release upon GitHub commit / PR merge.
- Live Repository: https://github.com/MD-Mushfiqur123/PayPal-AgentFlow